

Element B

Prior Attempt 1 – Invisible Door Lock:



Reviews say the mechanism allows the door to open by the width of an adult finger when locked which fails our criteria for the door to not move by more than 1 degree, but the mechanism is strong and the adhesive provides more than enough strength for its purpose, meeting our criteria for it to withstand 50N. It is stated to be too long for some cabinets that are especially small, but stalls are always at least human size. It's made from affordable materials, shown by its price tag at \$19, meeting our cost constraints. Promotion images show this locking mechanism relies on both a flush ceiling over the door, and a smooth door, which makes the solution unapplicable for our situation, as most stalls have a gap near the ceiling. Product descriptions and anecdotes from reviewers say this mechanism is also only applicable for scenarios in which the mechanism can be installed long term, as set-up and disassembly are told to not be quick, failing our time criteria, but succeeding at our criteria that it must keep the door closed for 4 minutes.

Prior Attempt 2 – Child Safety Lock:



This design uses extremely affordable materials, shown by its price tag at \$9 for 4 locks, and the reviewers report the double locks keep out any children who wish to open the lock, meeting our criteria that the lock must withstand 50N. The reviewers anecdote that the adhesive is moderately successful, but it fails to stick to surfaces which aren't prepped or especially smooth, which would include many stalls. The product images and description also tell that the lock can only be used when the edge of a door ends on a corner, so that one component of the mechanism can be placed at a different angle from the other component, making it only applicable to certain stalls. This mechanism also fails to meet the time constraints of setup in under 5 seconds because reviewers say it requires set-up and isn't made for repeated removal and application, so it can't be used for just using a public stall once.

Prior Attempt 3 – NestGuard Adhesive Door Lock:



This design uses affordable materials as well, shown by its \$18 price tag, meeting our cost constraints, but the chain and adhesive are not accessible for us, making the materials in this design not meet the material criteria that the design must use class materials. Reviewers report this lock is resistant to any human force, meeting our criteria for the lock to resist 50N of force, but the adhesive will fail from pulling if the surface it was applied on wasn't clean, and most bathroom stalls aren't clean. Product descriptions tell the design works for bathroom stalls geometry, and can be adjusted so the bathroom stall moves negligible amounts, but reviewers say the adhesive can't be removed and applied again continuously, not meeting our reusability criteria.

Citations:

Prior Design 1

Amazon.com, Inc. *Upgraded Invisible Baby Proofing Cabinet Latch Locks (10 Pack) – No Drilling or Tools Required for Installation, Works with Most Cabinets and Drawers, Works with Countertop Overhangs, Highly Secure.* Amazon, <https://www.amazon.com/Upgraded-Invisible-Proofing-Cabinet-Latch/dp/B07GRNTXJZ>. Accessed 14 Jan. 2026.

Prior Design 2

Amazon.com, Inc. *Child Safety Locks (4-Pack) — Adjustable Strap Length, Double Lock Option, Baby Proof Cabinets, Drawers, Toilets & Appliances.* Amazon,

<https://www.amazon.com/Child-Safety-Locks-4-Pack-Adjustable/dp/B07XR64L7P>. Accessed 14 Jan. 2026.

Prior Design 3

Amazon.com, Inc. *NestGuard Child Proof Door Lock with Adjustable Strap, Cat Door Latch 3M Adhesive Safety Locks for Baby, Dementia & Pets, No Tools Needed (2 Pack)*. Amazon,

<https://www.amazon.com/NestGuard-Adjustable-Adhesive-Safety-Dementia/dp/B0FLJYK4G6>. Accessed 14 Jan. 2026.